Which address is used by a layer-2 switch to make its forwarding decision?

a. IPv4 address
b. IPv6 address
c. MAC address
d. Port address

Which of the following command is used to set the device name on a router as CCLab1

a. Devicename CCLab1
b. Host Name Router CCLab1
c. Hostname CCLab1
d. Name CCLab1

Which one of the following methods is used to securely establish a remote CLI connection through a virtual interface, over a network?

a. Console
b. Telnet
c. AUX
d. SSH

Which of the following IPv6 migration techniques allows the existence of both the IPv4 and IPv6 addresses on the same network?

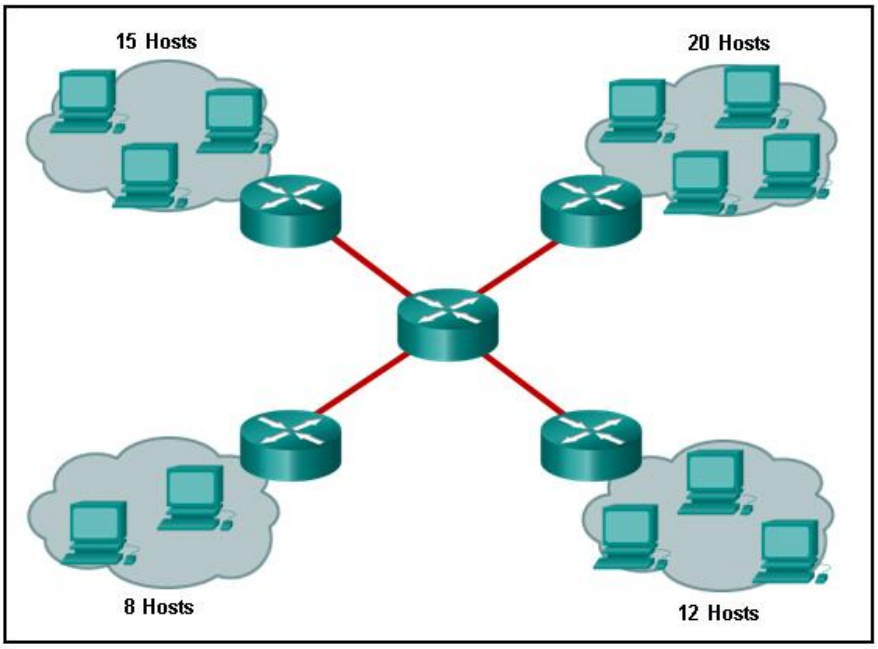
a. Tunnelling

b. Dual Stack

c. Translation

d. Encryption

Refer to the exhibit. How many broadcast domains are displayed?



- 1
- 4
- **8**

- 16
- 55

Which two characteristics describe the native VLAN? (Choose two.)

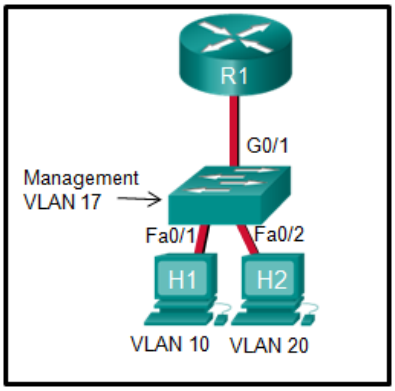
Designed to carry traffic that is generated by users, this type of VLAN is also known as the default VLAN.

- **The native VLAN traffic will be untagged across the trunk link.**
- This VLAN is necessary for remote management of a switch.
- High priority traffic, such as voice traffic, uses the native VLAN.
- **The native VLAN provides a common identifier to both ends of a trunk.**

On a switch that is configured with multiple VLANs, which command will remove only VLAN 100 from the switch?

- Switch# delete flash:vlan.dat
- Switch(config-if)# no switchport access vlan 100
- Switch(config-if)# no switchport trunk allowed vlan 100
- **Switch(config)# no vlan 100**

Refer to the exhibit. In what switch mode should port G0/1 be assigned if Cisco best practices are being used?

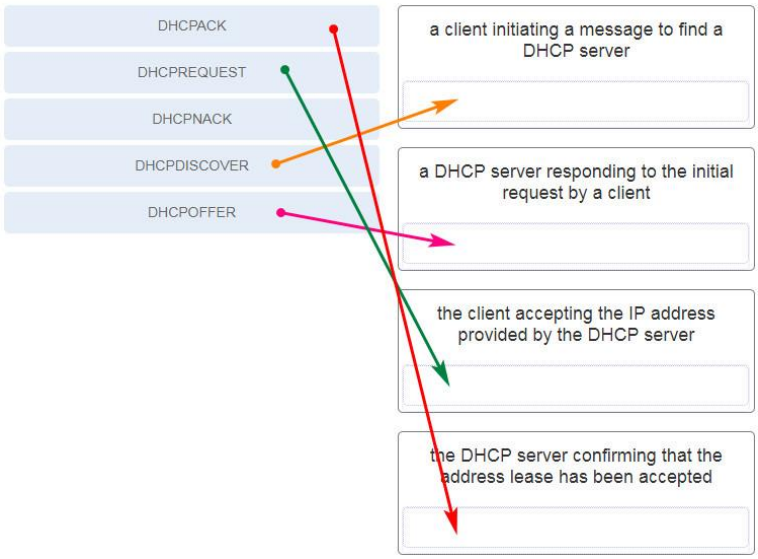


- access
- **trunk**
- native
- auto

A technician is configuring a new Cisco 2960 switch. What is the effect of issuing the BranchSw(config-if)# ipv6 address 2001:db8:a2b4:88::1/64 command?

- **It applies an IPv6 address to the virtual interface.**
- It activates a virtual or physical switch interface.
- It applies an IPv4 address to the virtual interface.
- It permits an IPv6 address to be configured on a switch physical interface.
- It updates the MAC address table for the associated port.

Match each DHCP message type with its description. (Not all options are used.)



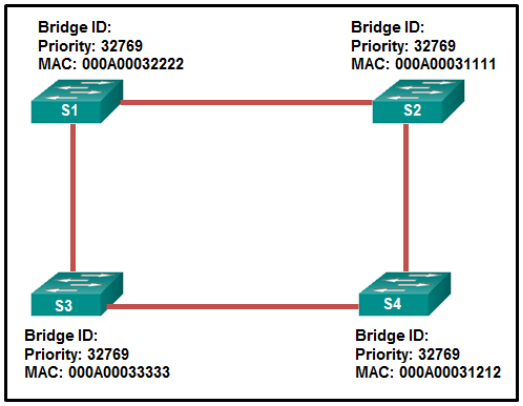
What additional information is contained in the 12-bit extended system ID of a BPDU?

- MAC address
- **VLAN ID**
- IP address
- port ID

In which two port states does a switch learn MAC addresses and process BPDUs in a PVST network? (Choose two.)

- disabled
- **forwarding**
- listening
- blocking
- **learning**

Refer to the exhibit. Which switch will be the root bridge after the election process is complete?



- S1
- **S2**
- S3
- S4

A network administrator is configuring an EtherChannel link between switches SW1 and SW2 by using the command SW1(config-if-range)# channel-group 1 mode auto. Which command must be used on SW2 to enable this EtherChannel?

- SW2(config-if-range)# channel-group 1 mode passive
- **SW2(config-if-range)# channel-group 1 mode desirable**
- SW2(config-if-range)# channel-group 1 mode on
- SW2(config-if-range)# channel-group 1 mode active

Which two network design features require Spanning Tree Protocol (STP) to ensure correct network operation? (Choose two.)

- static default routes
- implementing VLANs to contain broadcasts
- **redundant links between Layer 2 switches**
- link-state dynamic routing that provides redundant routes
- **removing single points of failure with multiple Layer 2 switches**

Refer to the exhibit. A network administrator is configuring a router for DHCPv6 operation. Which conclusion can be drawn based on the commands?

```
R1# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# ipv6 unicast-routing
R1(config)# ipv6 dhcp pool ACAD_CLASS
R1(config-dhcp)# dns-server 2001:db8:acad:a1::10
R1(config-dhcp)# domain-name netacad.net
R1(config-dhcp)# exit
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# ipv6 dhcp server ACAD_CLASS
R1(config-if)# ipv6 nd other-config-flag
R1(config-if)# end
R1#
```

- The router is configured for stateful DHCPv6 operation, but the DHCP pool configuration is incomplete.
- The DHCPv6 server name is ACAD_CLASS.
- Clients would configure the interface IDs above 0010.
- **The router is configured for stateless DHCPv6 operation.**

Which command will allow a network administrator to check the IP address that is assigned to a particular MAC address?

- Router# show running-config I section_dhcp
- Router# show ip dhcp server statistics
- **Router# show ip dhcp binding**
- Router# show ip dhcp pool

Which type of VLAN-hopping attack may be prevented by designating an unused VLAN as the native VLAN?

- DHCP spoofing
- DHCP starvation
- **VLAN double-tagging**
- DTP spoofing

Which access control component, implementation, or protocol is based upon usernames and passwords?

- 802.1X
- accounting
- **authentication**
- authorization

Which two Cisco solutions help prevent DHCP starvation attacks? (Choose two.)

- **DHCP Snooping**
- IP Source Guard
- Dynamic ARP Inspection
- **Port Security**

- Web Security Appliance

An IT security specialist enables port security on a switch port of a Cisco switch. What is the default violation mode in use until the switch port is configured to use a different violation mode?

- **shutdown**
- disabled
- restrict
- protect

Which access control component, implementation, or protocol collects and reports usage data?

- **accounting**
- authentication
- authorization
- 802.1X

What are two functions of dynamic routing protocols? (Choose two.)

- **to maintain routing tables**
- to assure low router overhead

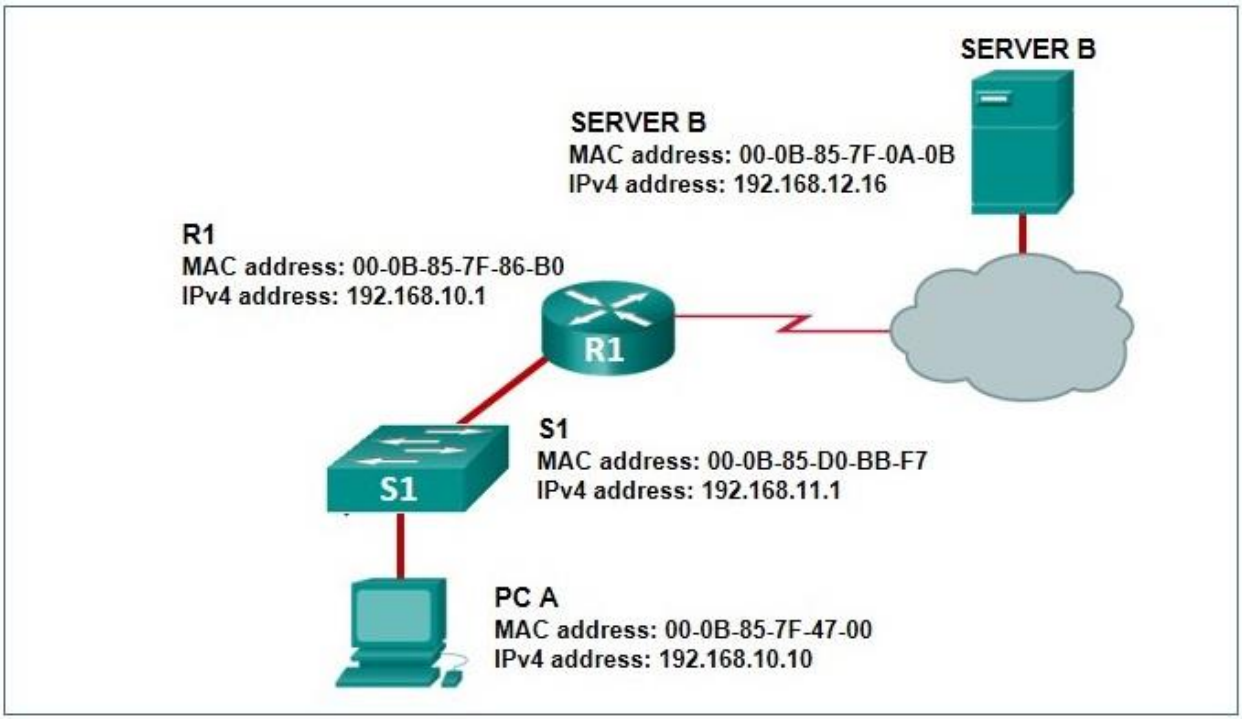
- to avoid exposing network information
- **to discover the network**
- to choose the path that is specified by the administrator

Refer to the exhibit. An administrator is attempting to install a default static route on router R1 to reach the Site B network on router R2. After entering the static route command, the route is still not showing up in the routing table of router R1. What is preventing the route from installing in the routing table?

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.0
R1#(config)# exit
R1# show ip route

<output omitted>

Gateway of last resort is not set

209.165.202.0/30 is subnetted, 1 subnets
C 209.165.202.128 is directly connected, Serial0/0/0

R1#
```

- The netmask is incorrect.
- The exit interface is missing.
- **The next hop address is incorrect.**
- The destination network is incorrect.

Refer to the exhibit. PC A sends a request to Server B. What IPv4 address is used in the destination field in the packet as the packet leaves PC A?



- 192.168.11.1
- 192.168.10.1
- **192.168.12.16**
- 192.168.10.10

When would it be more beneficial to use a dynamic routing protocol instead of static routing?

- in an organization where routers suffer from performance issues
- on a stub network that has a single exit point

- in an organization with a smaller network that is not expected to grow in size
- **on a network where there is a lot of topology changes**